

(the propensity score) and the weights are determined non-parametrically using local linear regression.³ This article, however, proposes a direct weighting scheme on the propensity score that can be used to estimate the effect of the treatment on the treated without estimating weights non-parametrically in a previous step.

Estimators of treatment effects that weight on functions of the probability of treatment are based on the Horvitz–Thompson statistic (Horvitz and Thompson, 1952). Non-parametric generalizations of the Horvitz–Thompson statistic have been studied in Imbens, Hirano and Ridder (2003). Ichimura and Linton (2002) have developed higher-order asymptotic expansions for this class of estimators to guide bandwidth selection. Robins and co-authors have proposed related estimators in the context of parametric models for time-varying treatments (see, *e.g.* Hernán, Brumback and Robins, 2001). In these studies it is assumed that all other factors, aside from the treatment, that affect the outcome variable are either observed, or their distribution is the same for treated and untreated. Consequently, all factors which confound simple comparisons of the outcome distribution between treated and untreated are observed. In contrast, the identification conditions used in this article allow for the distribution of both observed and unobserved factors to differ between treated and untreated, as long as the effect of unobserved factors on the outcome does not vary with time (or, more generally, if it experiences the same variation, on average, for treated and untreated).

3. NON-PARAMETRIC IDENTIFICATION

In the previous section, we referred to α as the “impact of the treatment” or the “treatment effect”, but the exact meaning of these terms was left undefined.

As in Rubin (1974) and Heckman (1990), the effect of the treatment will be defined in terms of potential outcomes. $Y^0(i, t)$ represents the outcome that individual i would attain at time t in absence of the treatment. In the same fashion, $Y^1(i, t)$ represents the outcome that individual i would attain at time t if exposed to the treatment. The effect of the treatment on the outcome for individual i at time t is then naturally defined as $Y^1(i, t) - Y^0(i, t)$.

The fundamental identification problem is that for any particular individual i and time period t , we do not observe both potential outcomes $Y^0(i, t)$ and $Y^1(i, t)$; so we cannot compute the individual treatment effect $Y^1(i, t) - Y^0(i, t)$. We only observe the realized outcome, $Y(i, t)$ that can be expressed as $Y(i, t) = Y^0(i, t) \cdot (1 - D(i, t)) + Y^1(i, t) \cdot D(i, t)$. Since, in the simple scenario considered here, the treatment is only administered after period $t = 0$, we can denote $D(i) = D(i, 1)$, then we have that $Y(i, 0) = Y^0(i, 0)$ and $Y(i, 1) = Y^0(i, 1) \cdot (1 - D(i)) + Y^1(i, 1) \cdot D(i)$.

Given the impossibility of computing individual treatment effects, researchers often focus on estimating some average effect, like the average effect of the treatment on the treated $E[Y^1(i, 1) - Y^0(i, 1) \mid D(i) = 1]$ (see, *e.g.* Heckman, 1990). Sometimes, when the desired level of aggregation is lower, researchers try to learn about some conditional version of the average effect on the treated $E[Y^1(i, 1) - Y^0(i, 1) \mid X(i), D(i) = 1]$.

For the rest of the article, the individual argument i will be dropped to reduce notation. I will take the next assumption to hold throughout this article.

Assumption 3.1. $E[Y^0(1) - Y^0(0) \mid X, D = 1] = E[Y^0(1) - Y^0(0) \mid X, D = 0]$.

Assumption 3.1 is the crucial identifying restriction in DID models. It states that, conditional on the covariates, the average outcomes for treated and controls would have followed parallel

3. Blundell *et al.* (2001) propose a related estimator which combines DID and matching on the propensity score.

paths in absence of the treatment.⁴ Notice that when $E[Y^0(0) | X, D = 1] = E[Y^0(0) | X, D = 0]$, Assumption 3.1 collapses to a “selection on observables” restriction ($E[Y^0(1) | X, D = 1] = E[Y^0(1) | X, D = 0]$) which can be used in cross-sectional studies to identify the effect of the treatment on the treated.⁵ Therefore, the results in this article also apply to this particular situation. This is the case for Ashenfelter’s dip example, discussed in Section 2. I will come back to this point in Section 3.2.

Existence of the expectations is assumed throughout. Under Assumption 3.1, the effect of the treatment on the treated conditional on X can be expressed as (Heckman *et al.*, 1997):⁶

$$E[Y^1(1) - Y^0(1) | X, D = 1] = \{E[Y(1) | X, D = 1] - E[Y(1) | X, D = 0]\} - \{E[Y(0) | X, D = 1] - E[Y(0) | X, D = 0]\}. \quad (9)$$

Even when Assumption 3.1 holds unconditionally, if it also holds conditional on some predetermined covariates of interest (*e.g.* gender), we may still use the conditional identification result to evaluate the effect of the treatment for different groups of the population (*e.g.* women vs. men).

In principle, the identification result in equation (9) can be used to estimate $E[Y^1(1) - Y^0(1) | X, D = 1]$ by producing non-parametric estimates of each one of the four expectations on the R.H.S. of equation (9). In practice, the number of observations required to attain an acceptable precision for this type of non-parametric estimator increases very rapidly with the dimension of X . This problem, often called the curse of dimensionality, may prevent us from using non-parametric estimators for $E[Y^1(1) - Y^0(1) | X, D = 1]$ in many practical instances. In addition, a simple non-parametric estimator of $E[Y^1(1) - Y^0(1) | X, D = 1]$, directly based on equation (9), may be difficult to interpret if the dimension of X is larger than two, so we cannot summarize the result in a graph. In those cases, some integration over the distribution of X for the treated is required in order to produce summary statistics. Even then, the estimation process is cumbersome. An estimator of average treatment effects for the treated directly based on equation (9) requires estimating four conditional expectations non-parametrically (or two if a sample with repeated outcomes, $Y(0)$ and $Y(1)$, for the same individuals is available) and then integrating the estimates to the desired level of aggregation.

This article proposes simple weighting schemes to produce estimators of the average effect on the treated $E[Y^1(1) - Y^0(1) | D = 1]$ and parsimonious parametric approximations to its conditional version $E[Y^1(1) - Y^0(1) | X_k, D = 1]$, where X_k is a function of X (for example,

4. Using experimental data, Heckman *et al.* (1997, 1998) have shown the plausibility of this identifying assumption in the context of the evaluation of a subsidized training programme.

5. In that case, Assumption 3.1 implies $E[Y^1(1) - Y^0(1) | X, D = 1] = E[Y(1) | X, D = 1] - E[Y(1) | X, D = 0]$, and pre-treatment data are not required to identify the average effect of the treatment on the treated. If, in addition, $E[Y^1(1) | X, D = 1] = E[Y^1(1) | X, D = 0]$, then $E[Y^1(1) - Y^0(1) | X] = E[Y(1) | X, D = 1] - E[Y(1) | X, D = 0]$. Selection on observables implies that all factors which confound simple comparisons of outcomes between treated and controls are observed. This is a too stringent assumption if the distribution of unobserved variables which affect the outcome is believed to differ between treated and controls. See, *e.g.* Rubin (1977) and Heckman *et al.* (1997).

6. A researcher could be interested in estimating $E[Y^1(1) | X, D = 1]$ and $E[Y^0(1) | X, D = 1]$ separately. Since $Y^1(1)$ is observed for the treated, both conditional expectations are identified. In fact, given that $Y^1(1)$ is observed for the treated, identification results on $E[Y^1(1) - Y^0(1) | X, D = 1]$ are equivalent to identification results on $E[Y^0(1) | X, D = 1]$. Here, I concentrate on the difference $E[Y^1(1) - Y^0(1) | X, D = 1]$ because it has been the object of interest in the difference-in-differences literature (see, *e.g.* Heckman *et al.* (1997, 1998)). Notice also that Assumption 3.1 by itself does not identify the average treatment effects conditional only on the covariates ($E[Y^1(1) - Y^0(1) | X]$), unless conditional average effects coincide for treated and untreated $E[Y^1(1) - Y^0(1) | X, D = 1] = E[Y^1(1) - Y^0(1) | X, D = 0]$. The reason is that Assumption 3.1 identifies $E[Y^0(1) | X, D = 1]$, and therefore, the effect of the treatment on the treated. However, Assumption 3.1 leaves $E[Y^1(1) | X, D = 0]$ totally unrestricted; so the effect of the treatment on the untreated is also unrestricted.

a subset of the variables in X). The weighting scheme is directly based on the propensity score, $P(D = 1 | X)$, which is the only function which needs to be estimated in a first step. As a result, the proposed method reduces the first step estimation burden and allows the researcher to use four or two times more observations for first step estimation, relative to direct estimation of equation (9). In practice, this feature may be an important advantage if non-parametric estimation is carried out in the first step. When the number of observations is too small for non-parametric estimation in the first step, the proposed method allows the researcher to circumvent the curse of dimensionality by placing parametric restrictions on the propensity score, which leaves $E[Y^1(1) - Y^0(1) | X_k, D = 1]$ unrestricted, rather than on each one of the conditional means of equation (9), which may impose unwanted restrictions on $E[Y^1(1) - Y^0(1) | X_k, D = 1]$.

The estimation of parametric approximations to $E[Y^1(1) - Y^0(1) | X_k, D = 1]$ has some benefits. First, it provides a simple method to produce estimation results at the level of aggregation desired by the analyst. In addition, the results are parsimoniously summarized by the estimates of the parameters that define the approximation to $E[Y^1(1) - Y^0(1) | X_k, D = 1]$. However, the quality of the information provided by our estimators will be low if the quality of the approximation to $E[Y^1(1) - Y^0(1) | X_k, D = 1]$ is poor.

Since the object of study is the effect of the treatment on the treated, the minimal requirement for the problem to be well defined is that some fraction of the population is exposed to the treatment. In addition, since identification is attained after controlling for the effect of some covariates X , it will be required that for each given value of the covariates there is some fraction of the population that remains untreated and can be used as controls.

Assumption 3.2. $P(D = 1) > 0$ and with probability one $P(D = 1 | X) < 1$.

Note that Assumption 3.2 implies that the support of the propensity score for the treated is a subset of the support of the propensity score for the untreated. This is a well-known condition for identification of the average impact on the treated under selection on covariates (see, e.g. Heckman *et al.*, 1997).

3.1. Random sample with repeated outcomes

In this section, I introduce the identification strategy proposed in this article by considering the situation in which we can observe both pre-treatment and post-treatment outcomes for a random sample of the population of interest. Examples of applications of DID estimators to data on repeated outcomes are Card and Krueger (1994), Heckman *et al.* (1997, 1998), Garvey and Hanka (1999) and Blundell *et al.* (2001). Under this sampling scheme, for each individual in our sample we observe $(Y(1), Y(0), D, X)$. Later in the article, the identification procedure is extended to repeated cross sections.

Lemma 3.1. *If Assumption 3.1 holds, and for values of X such that $0 < P(D = 1 | X) < 1$, we have $E[Y^1(1) - Y^0(1) | X, D = 1] = E[\rho_0 \cdot (Y(1) - Y(0)) | X]$, where*

$$\rho_0 = \frac{D - P(D = 1 | X)}{P(D = 1 | X) \cdot (1 - P(D = 1 | X))}.$$

For notational convenience, let $\rho_0 = -1$ if $P(D = 1 | X) = 0$ (this choice is inconsequential since the objects of interest will be integrals over the distribution of the X conditional on $D = 1$). The average effect of the treatment for the treated is given by

$$\begin{aligned}
E[Y^1(1) - Y^0(1) \mid D = 1] &= \int E[Y^1(1) - Y^0(1) \mid X, D = 1] dP(X \mid D = 1) \\
&= \int E[\rho_0 \cdot (Y(1) - Y(0)) \mid X] dP(X \mid D = 1) \\
&= E \left[\rho_0 \cdot (Y(1) - Y(0)) \cdot \frac{P(D = 1 \mid X)}{P(D = 1)} \right] \\
&= E \left[\frac{Y(1) - Y(0)}{P(D = 1)} \cdot \frac{D - P(D = 1 \mid X)}{1 - P(D = 1 \mid X)} \right]. \tag{10}
\end{aligned}$$

In words, under Assumptions 3.1 and 3.2, a simple weighted average of temporal differences in the outcome variable recovers the average effect of the treatment for the treated. The weights depend on the propensity score. On an intuitive level, this scheme works by weighting-down the distribution of $Y(1) - Y(0)$ for the untreated for those values of the covariates which are over-represented among the untreated (that is, with low $P(D = 1 \mid X)/P(D = 0 \mid X)$), and weighting-up $Y(1) - Y(0)$ for those values of the covariates under-represented among the untreated (that is with high $P(D = 1 \mid X)/P(D = 0 \mid X)$). In this way the same distribution of the covariates is imposed for treated and untreated.⁷

Equation (10) suggests a simple two-step method to estimate the average effect of the treatment on the treated under Assumptions 3.1 and 3.2: (i) estimate the propensity score, $P(D = 1 \mid X)$, and compute the fitted values for the sample; (ii) plug the fitted values into the sample analogue of equation (10) to obtain an estimate of $E[Y^1(1) - Y^0(1) \mid D = 1]$.

In many practical instances, the desired level of aggregation is lower than the entire treated population and the analyst wants to study how the treatment affects the treated for different groups of the population. As explained above, a non-parametric estimator of $E[Y^1(1) - Y^0(1) \mid X, D = 1]$ may be difficult to interpret, especially if the dimension of X is large. Such a problem is circumvented here by focusing on parametric approximations to $E[Y^1(1) - Y^0(1) \mid X, D = 1]$. More generally, consider the situation in which we need to condition on some vector of random variables X to attain identification, but we are interested in $E[Y^1(1) - Y^0(1) \mid X_k, D = 1]$, where X_k is some deterministic function of X .⁸ This situation is relevant when the number of covariates needed in order to attain identification is large, so the analyst may be willing to allow for a higher level of aggregation in the second step in order to obtain parsimonious results.

Consider a class of approximating functions $\mathcal{G} = \{g(X_k; \theta) : \theta \in \Theta \subset \mathbb{R}^k\}$, square-integrable with respect to $P(X_k \mid D = 1)$. Then, a least squares approximation from $\mathcal{G}$ to $E[Y^1(1) - Y^0(1) \mid X_k, D = 1]$ is given by $g(X_k; \theta_0)$ where

$$\theta_0 = \arg \min_{\theta \in \Theta} E[\{E[Y^1(1) - Y^0(1) \mid X_k, D = 1] - g(X_k; \theta)\}^2 \mid D = 1]. \tag{11}$$

For example, if $\mathcal{G} = \{X_k' \theta : \theta \in \Theta \subset \mathbb{R}^k\}$, then θ_0 defines a linear least squares approximation to $E[Y^1(1) - Y^0(1) \mid X_k, D = 1]$. It is assumed that θ_0 exists uniquely.

Proposition 3.1. *If Assumptions 3.1 and 3.2 hold, then*

$$\theta_0 = \arg \min_{\theta \in \Theta} E[P(D = 1 \mid X) \cdot \{\rho_0 \cdot (Y(1) - Y(0)) - g(X_k; \theta)\}^2].$$

7. Similarly, Heckman *et al.* (1997) use matching on the propensity score to account for imbalances in the distribution of the covariates between treated and untreated. Matching on the propensity score works because it imposes the same distribution of the covariates for the two groups (see Rosenbaum and Rubin, 1983).

8. For example, X_k may contain a subset of the variables in X . In other instances, X may contain indicators for all different values of a discrete variable included in X_k .